Is all well with your models?: Strategies To Deal With Concept Drifts

A case of spandex fiber production

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ABSTRACT

- This poster presents a concept drift detection methods that can give an insight into the situation when the performance of the model deteriorates due to concept drift, auto-modeling may not be carried out effectively.
- We adopt **feature selection** and various **model update strategies** into the baseline algorithm, which is conventional AL Framework, to detect the concept drift effectively and efficiently.
- These advanced strategies could successfully detect concept drift in real-world scenarios, ensure consistent performance, and reliable predictions in dynamic environments.

INTRODUCTION

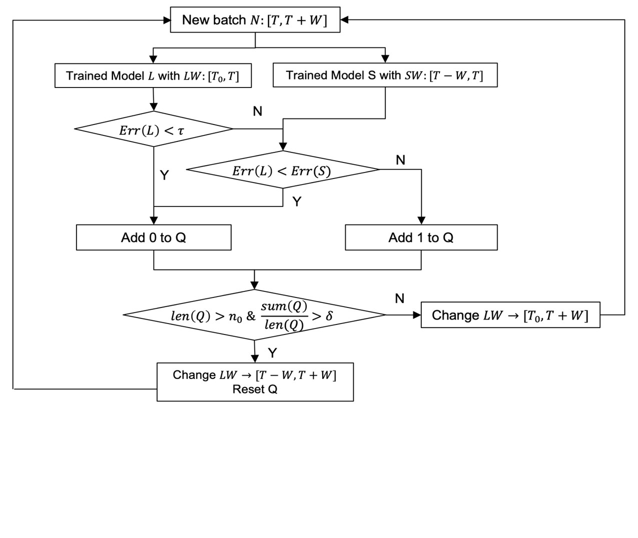
- **Concept drift** refers to the situation where the statistical properties of the target variable, which the model aims to predict, change unpredictably over time. When concept drift occurs, the patterns learned from past data may become irrelevant to new data, resulting in inaccurate predictions and decision outcomes.
- This phenomenon is prevalent in many **real-world** scenarios, highlighting the need for appropriate **model update strategies** to address concept drift.
- We aim to introduce more specific model update strategies, focusing on the methodology of concept drift detection.

KEY FINDINGS

- Conducted research on the concept drift as a necessary concept for establishing an effective **Auto-modeling system.**
- Proposed a specific strategies for responding to **concept drift detection.**
- Use the concepts of **'period'** and **'weight'** to specify a **model update strategy**.
- Applied specific model update strategy to **'real-world'** situations, Korean Spandex Industry.

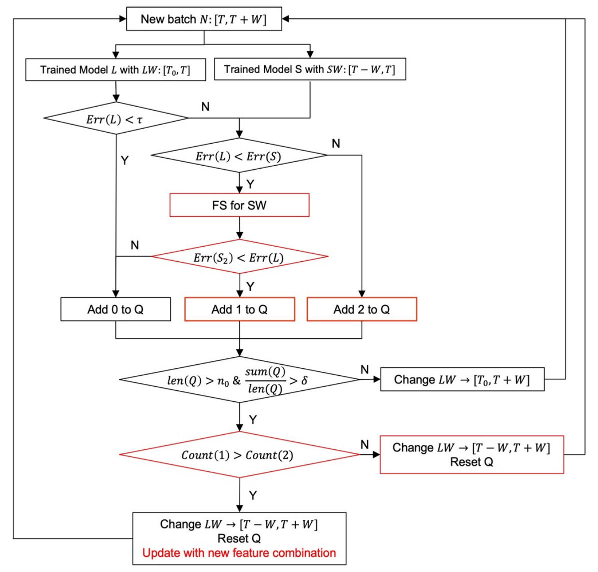
METHOD 1 – ADVANCD AL FRAMEWORK

Conventional AL Framework



- Train the model with cumulative data long window(LW) and recent short window(SW) separately.
- **Compare the errors** for N to detect CD.
- When CD is detected, replace LW with SW + N.

Advanced AL Framework



- An advanced method that improves the limitations of AL.
- Existing AL **limitations**: Dealing with concept drift by adjusting the learning interval and updating the model without modifying the model itself.
- Proposed improvement: Establishing a model update strategy that includes **feature selection (FS).**

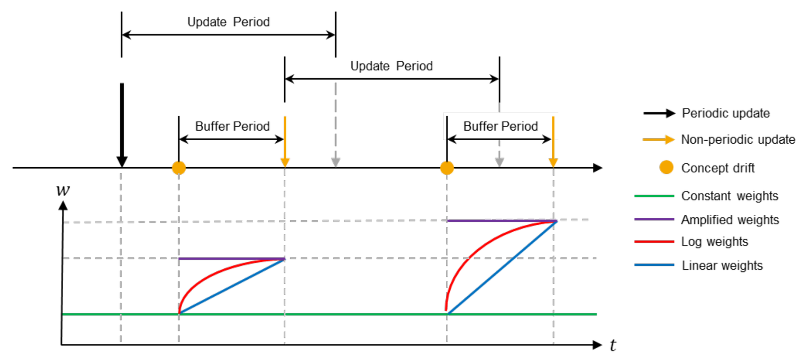
METHOD 2 – MODEL UPDATE STRATEGY

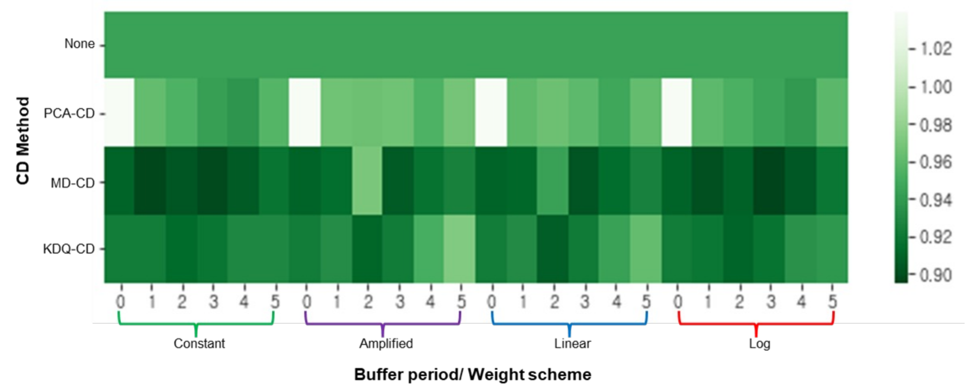
- Relying solely on hyperparameters in the original AL framework is **insufficient** to achieve improved concept drift detection results.
- When concept drift is either detected too frequently or the detection is overly sensitive to user-specified hyperparameters, there is a need for model update strategies to adjust and address these issues.

- **Period**: Definition of the period for establishing CD response strategies.

Type	Explanation
Reference Period	A past period of similar duration used for diagnosing pattern changes and model updates.
Update Period	The minimum duration required to accumulate new data for model updates.
Buffer Period	The period during which additional data is gathered after detecting pattern changes until model updates are performed.

- **Differential Weighting**: Weight assigned to the buffer period.





REFERENCE AND ACKNOWLEDGMENT

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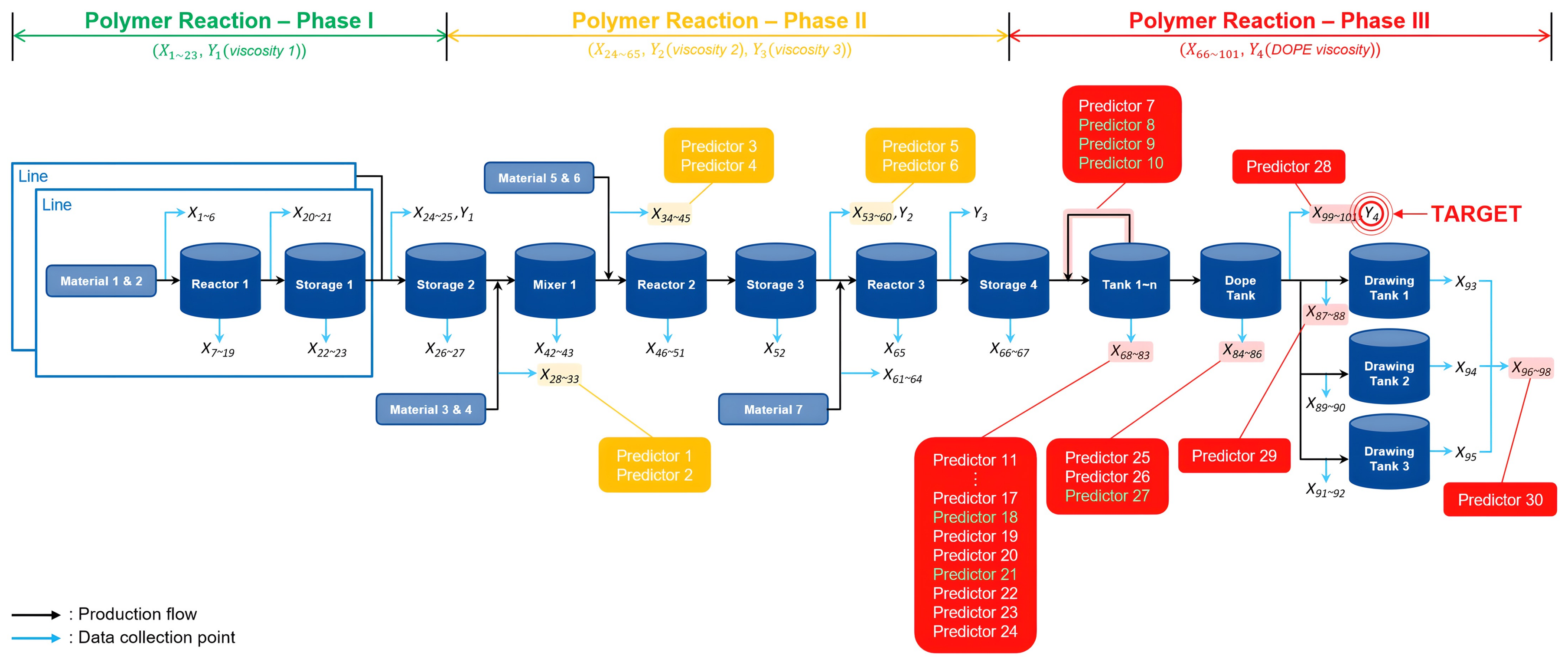
APPLICATION

Spandex Industry



Experimental Design

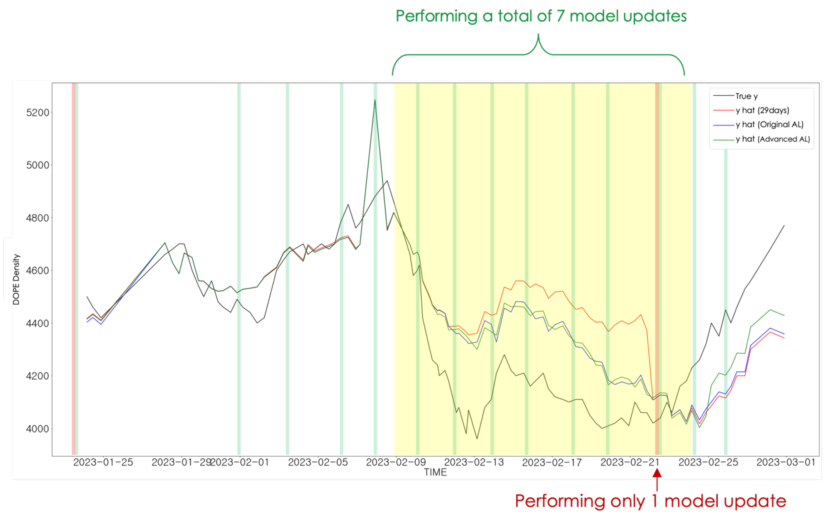
- Datapoints – 691,200 (20.07~21.12)
- Base model – Lasso Regression
- Target Variable – Dope Polymer Viscosity
- Prediction – 30 features (24 process features + 6 derived features)
- Validation – Rolling origin with the window size of 15 days and a test period of 5 days

Continuous Process in Spandex fiber production

- The manufacturing of spandex fibers can be divided into three reaction phases: pre-polymer production, polymer reactions, and fiber production.
- Must be subjected to a variety of chemical tests to obtain quality indicators.
- Hard to optimize the process in real-time.
- Hard to make constant flow time of intermediate products.

Comparative Results of Model Update Strategies

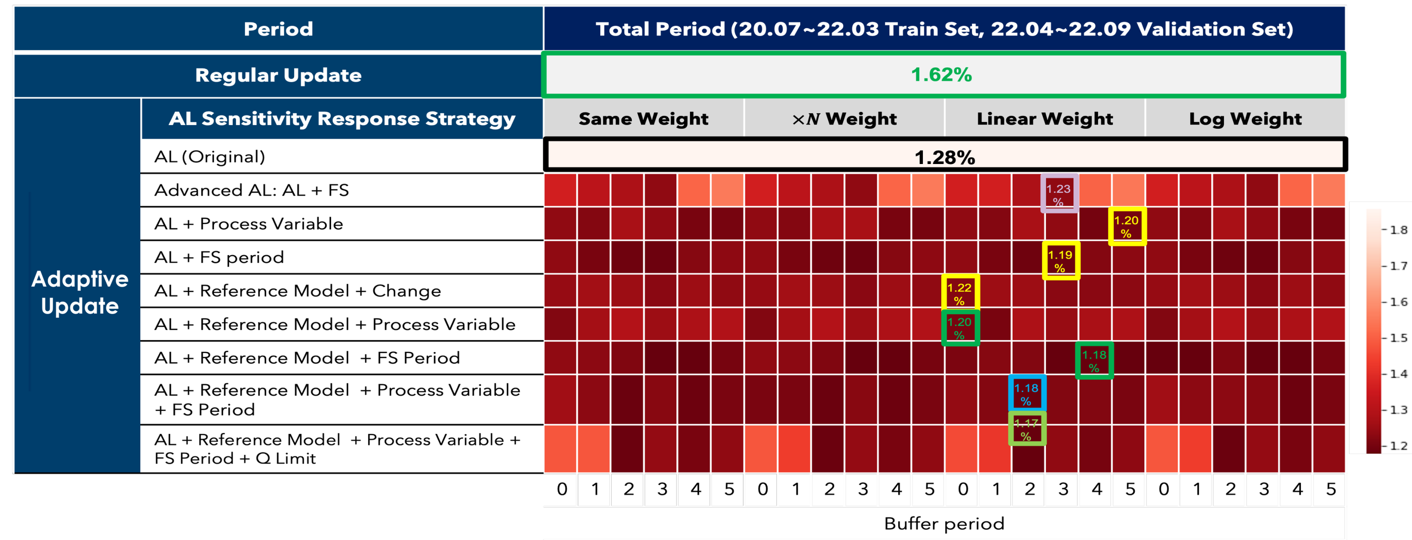
- Detecting downturn periods and performing model updates sensitively to the latest data.
- Detecting rebound periods after vulnerable periods and performing model updates.



Comparison of Predictive Performance (MAPE)

Type of Strategy	Overall Period (21.09.01~23.02.28)	Rapid Change Period (23.02.09~23.02.24)
Periodic model update strategy	6.48%	**2.94%**
AL + Adaptation strategy	4.88%	**2.45%**
Auto-modeling strategy	4.69%	**1.73%**

Ablation Study



- Optimal Auto-Modeling Strategy: AL + Reference Model + Process Variable + FS Period + Q Limit *(MAPE=1.17%)*

1) Reference Model: Regular Model with 29 days
2) Process Variable: Included
3) FS Period: **4 days**
4) Q Length: **11 days**
5) Buffer Period: **2 days**
6) Weight: **Linear**